3D CNN (cnn3d_final, best model): convolutions over hours and cells, no recurrence

1 Input x (one 12 h period)

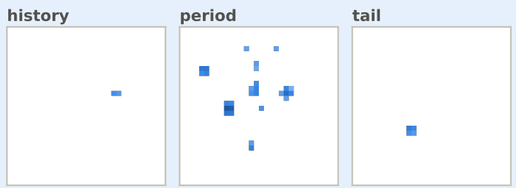
24 hourly steps = max of four 15-min slots
6 h history · 12 h period · 6 h tail

7 channels per step and cell

frp	log(1 + FRP) / log(1 + 1100)
confidence	confidence / 100
detected	1 if detected in the hour
period	1 inside the 12 h period
time	step / 23
pt	mainland Portugal mask
persistent	off-season pixel flag

shape 24 × 7 × H × W
training: H = W = 32 (crops)
inference: 177 × 106 (full grid)

example crop (32 × 32 cells): max frp per part



2 3D CNN (cnn3d_final): 63,073 parameters

Conv3d 3×3×3
7 → 16 channels, ReLU
3 hours × 3 × 3 cells per output
→ 16 × 24 × H × W

Conv3d, stride 2 in t
16 → 32 channels, ReLU
time 24 → 12 steps
→ 32 × 12 × H × W

Conv3d, stride 2 in t
32 → 32 channels, ReLU
time 12 → 6 steps
→ 32 × 6 × H × W

Pool over time
max and mean of the 6 steps, concatenated
→ 64 × H × W

What each output step sees (receptive field of the three Conv3d layers):

space: 3 → 5 → 7 cells (9 × 9 after the head) time: 3 → 5 → 9 hours, sampled every 4 hours

All hours enter together; there is no state carried from hour to hour. Max and mean over the 6 steps then combine the whole 24 h window: the maximum keeps the strongest event, the mean how long it lasted.

trained with 2 hard negatives per positive

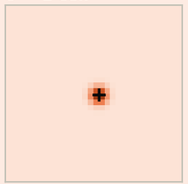
3 Head

Conv 3×3 → 32, ReLU
Conv 1×1 → 1 logit
sigmoid → p

p = probability that an ICNF fire ≥ 50 ha started in this cell during the 12 h period

4 Target y

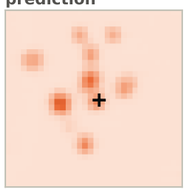
σ = 1 cell



focal loss vs. p

5 Output p

prediction



3 × 3 peaks, p ≥ 0.05
→ sub-pixel points
(lat, lon, score)

example: 2016-08-06
06–18 UTC (validation)
+ = target ignition